

Biased Collective-Coordinate Potential for Hole-Size Measurement

CCO-projects — `biased_optimization` branch

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1 Original collective-coordinate potential

The standard stealthy collective-coordinate (CC) potential is

$$\Phi_{\text{CC}}(\{\mathbf{r}_i\}) = \frac{1}{V} \sum_{\mathbf{k} \in \mathcal{K}} \tilde{v}(\mathbf{k}) |\rho(\mathbf{k})|^2, \quad (1)$$

where

$$\rho(\mathbf{k}) = \sum_{j=1}^N e^{i\mathbf{k} \cdot \mathbf{r}_j} \quad (\text{structure-factor amplitude}), \quad (2)$$

$$\mathcal{K} = \bigcup_{n=1}^M \{\mathbf{k} : k_1^{(n)} < |\mathbf{k}| \leq k_1^{(n)} + \delta^{(n)}\} \quad (\text{excluded } k\text{-space region}), \quad (3)$$

V is the simulation-box volume, and $\tilde{v}(\mathbf{k}) = 1$ for the flat (default) stealthy weight. Minimising Φ_{CC} to zero enforces $S(\mathbf{k}) = 0$ for all $\mathbf{k} \in \mathcal{K}$, the defining condition of stealthy hyperuniformity.

2 Goal: maximum hole radius at the box centre

We wish to measure the maximum radius R_c of a spherical void that can be placed at the box centre

$$\mathbf{r}_c = \frac{1}{2}(\mathbf{a}_1 + \mathbf{a}_2 + \cdots + \mathbf{a}_d), \quad (4)$$

where $\mathbf{a}_i$ are the lattice vectors, while the configuration simultaneously satisfies the stealthy constraint $\Phi_{CC} \approx 0$.

3 Exclusion field added to the potential

To enforce a void of probe radius R_f around $\mathbf{r}_c$, we add a radial exclusion field that diverges as any particle approaches the centre:

$$\phi_{\text{ex}}(\{\mathbf{r}_i\}; R_f) = \sum_{\substack{i=1 \\ r_{ic} < R_f}}^N \left(\frac{R_f}{r_{ic}} - 1 \right), \quad r_{ic} = |\mathbf{r}_i - \mathbf{r}_c|. \quad (5)$$

Key properties of ϕ_{ex} :

- It is *zero* at $r_{ic} = R_f$ (particle on the boundary).
- It *diverges* as $r_{ic} \rightarrow 0$ (particle at the centre).
- It is *non-negative* for all $r_{ic} > 0$.
- Only particles strictly inside the probe sphere ($r_{ic} < R_f$) contribute.

4 Biased potential

The full biased potential used during minimisation is

$$\Phi_{\text{biased}}(\{\mathbf{r}_i\}; R_f) = \underbrace{\frac{1}{V} \sum_{\mathbf{k} \in \mathcal{K}} \tilde{v}(\mathbf{k}) |\rho(\mathbf{k})|^2}_{\Phi_{CC}} + \underbrace{\sum_{\substack{i \\ r_{ic} < R_f}} \left(\frac{R_f}{r_{ic}} - 1 \right)}_{\phi_{\text{ex}}}. \quad (6)$$

When Φ_{biased} is minimised to a value below a threshold ε_{tol} , both conditions are jointly satisfied:

1. The stealthy constraint $S(\mathbf{k}) \approx 0$ holds for all $\mathbf{k} \in \mathcal{K}$ (from $\Phi_{CC} \approx 0$).
2. All particles lie outside the sphere of radius R_f centred at $\mathbf{r}_c$ (from $\phi_{\text{ex}} \approx 0$, which requires $r_{ic} \geq R_f$ for every particle).

5 Force from the exclusion field

The force on particle i from the exclusion field is

$$\mathbf{F}_i^{\text{ex}} = -\nabla_{\mathbf{r}_i} \phi_{\text{ex}} = \begin{cases} \frac{R_f}{r_{ic}^3} (\mathbf{r}_i - \mathbf{r}_c) & \text{if } r_{ic} < R_f, \\ \mathbf{0} & \text{otherwise.} \end{cases} \quad (7)$$

This force points *radially outward* from $\mathbf{r}_c$ and grows without bound as $r_{ic} \rightarrow 0$, acting as a hard barrier that prevents any particle from entering the probe sphere.

The total force on particle i during minimisation is

$$\mathbf{F}_i = \mathbf{F}_i^{\text{CC}} + \mathbf{F}_i^{\text{ex}}, \quad (8)$$

where $\mathbf{F}_i^{\text{CC}} = -\nabla_{\mathbf{r}_i} \Phi_{\text{CC}}$ is the standard stealthy force already present in the codebase.

6 Feasibility scan and maximum hole radius

The probe radius R_f is treated as a fixed external parameter that is stepped upward from $R_{\min}$ to $R_{\max}$ in increments ΔR :

$$R_f \in \{R_{\min}, R_{\min} + \Delta R, R_{\min} + 2\Delta R, \dots\}. \quad (9)$$

At each value of R_f :

1. N_{trial} independent random initial configurations are generated and minimised under Φ_{biased} .
2. If *any* minimisation reaches $\Phi_{\text{biased}} < \varepsilon_{\text{tol}}$, the void of radius R_f is **feasible**: record $R_c = R_f$ and advance to the next step.
3. If *all* N_{trial} attempts fail, the void is **infeasible**: stop the scan.

The output is the maximum feasible hole radius R_c and the corresponding stealthy configuration.

7 Dimensionless output

Results are reported in two units:

- R_c in absolute (simulation) length units.
- $R_c \cdot k_{\max}$ (dimensionless), where $k_{\max}$ is the outer edge of the outermost excluded shell:

$$k_{\max} = \max_n (k_1^{(n)} + \delta^{(n)}). \quad (10)$$

Running the scan across many independent seeds (SLURM array jobs, each with a different random initial condition) samples the distribution of R_c over the stealthy ground-state ensemble, giving the hole-size distribution $p(R_c)$.